Voice-Based Concept Analyser

AI-Powered Semantic Understanding and Speech Fluency Evaluation System

1. Project Title

Voice-Based Concept Analyser: AI-Powered Semantic Understanding and Speech Fluency Evaluation System

2. Abstract

The Voice-Based Concept Analyser is an AI-powered application designed to evaluate a learner's conceptual understanding and communication skills through spoken explanations.

The user records or uploads an audio explanation of a technical concept such as Machine Learning, Artificial Intelligence, or Cloud Computing. The system converts speech into text using the Whisper speech recognition model and evaluates semantic meaning by comparing the explanation with a predefined reference concept using Sentence-BERT embeddings and cosine similarity.

In addition to conceptual evaluation, the system analyses speech fluency using audio processing techniques. It measures filler word usage, pause ratio, RMS energy, hesitation patterns, and speech clarity. Results are presented through an interactive Streamlit dashboard.

The system also generates a structured PDF report containing transcription, semantic similarity score, speech analysis metrics, waveform visualization, comprehension score, and qualitative feedback. The application is useful for interview preparation, viva examinations, presentations, seminars, and technical assessments.

3. Problem Statement

Students often find it difficult to evaluate whether their spoken explanation of a concept is technically correct, complete, and clearly communicated. Traditional assessment methods mainly depend on manual evaluation by teachers or interviewers, which can be time-consuming and may not provide immediate feedback.

The Voice-Based Concept Analyser addresses this problem by combining speech recognition, natural language processing, semantic similarity analysis, and audio signal processing. It evaluates both conceptual correctness and communication fluency, then provides numerical scores, visual analysis, and qualitative feedback.

4. Features

Speech-to-Text Transcription

Converts recorded or uploaded audio explanations into text using Whisper.

Semantic Understanding Evaluation

Compares the transcription with a predefined reference explanation using Sentence-BERT embeddings and cosine similarity.

Concept Coverage Analysis

Identifies correctly covered ideas, missing concepts, conceptual gaps, and possible topic deviation.

Speech Fluency Analysis

Evaluates filler word count, filler word frequency, pause ratio, speech duration, hesitation patterns, and speaking clarity.

Audio Energy Analysis

Uses Librosa-based processing to calculate RMS energy and analyse voice consistency.

Waveform Visualization

Generates waveform representations to visualize speech intensity and silence patterns.

Comprehension Score

Combines conceptual and speech-related metrics to classify performance as Strong Understanding, Moderate Understanding, or Poor Understanding.

Interactive Dashboard

Displays audio playback, transcription, semantic similarity, concept coverage, fluency metrics, waveform visualization, final score, and qualitative feedback.

PDF Report Generation

Creates a downloadable report containing topic details, transcription, evaluation metrics, waveform image, final score, and feedback.

5. Tech Stack

Programming Language: Python

Frontend and User Interface: Streamlit

Speech Recognition: Whisper

Natural Language Processing: Sentence-BERT, Sentence Transformers, Transformers

Semantic Evaluation: Sentence Embeddings, Cosine Similarity, Scikit-learn

Audio Processing: Librosa, SoundFile, NumPy

Data Processing: Pandas, NumPy

Visualization: Matplotlib

PDF Report Generation: ReportLab

Development and Version Control: Git, GitHub

6. Installation Steps

Step 1 – Clone the Repository: Clone the GitHub repository and move into the project directory.

Step 2 – Create a Virtual Environment: Use `python -m venv venv` and activate it.

Step 3 – Install Dependencies: Run `pip install -r requirements.txt`.

Step 4 – Run the Application: Run `streamlit run app.py`. The application will open in the browser or provide a local address in the terminal.

7. Usage Instructions

- Start the application using Streamlit.
- Select a concept or topic for evaluation.
- Record or upload an audio file containing your explanation.
- Submit the audio for analysis.
- The system transcribes speech using Whisper.

- The transcription is compared with the reference explanation using Sentence-BERT semantic embeddings.
- The application calculates semantic similarity and analyses concept coverage.
- The audio is processed to measure filler words, pauses, RMS energy, and other fluency metrics.
- Review the complete results in the dashboard.
- Download the generated PDF report for future reference or progress tracking.

8. Screenshots

Screenshots can be added to the repository to demonstrate the application interface and analysis results.

- Home page and audio upload section
- Transcription and semantic similarity results
- Speech fluency analysis with filler word statistics, pause ratio, and RMS energy
- Audio waveform visualization
- Final comprehension score and qualitative feedback
- Sample generated PDF report

9. Future Scope

- Real-time speech analysis while the user is speaking.
- Support for multiple languages.
- Automatic generation of reference concepts.
- AI-generated personalized improvement recommendations.
- User authentication and individual profiles.
- Database integration for storing previous analysis results.
- Historical performance tracking and progress dashboards.
- Teacher and student dashboards.
- Automatic question generation based on selected topics.
- Integration with Learning Management Systems.
- Mobile application development.
- Cloud deployment for public access.
- Real-time mock interview simulation.
- Dynamic follow-up questions based on user answers.
- Improved pronunciation analysis.
- Speech emotion and confidence estimation.
- Support for multiple reference answers for more flexible evaluation.

Conclusion

The Voice-Based Concept Analyser provides an automated approach to evaluating both conceptual understanding and spoken communication.

By combining speech recognition, semantic similarity analysis, natural language processing, and audio signal analysis, the system provides learners with detailed feedback on what they explain and how effectively they communicate it.

The project can support interview preparation, viva practice, academic presentations, technical training, and self-assessment. Its combination of semantic evaluation, fluency analysis, visualization, and downloadable reports makes it a useful educational assessment and learning support tool.